Unified Substrate Theory (UST)

A New Framework for Everything

Plain English Edition

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Written for anyone who has ever asked: "What is the universe actually made of?"

No physics background required. Every technical section has a Plain English translation.

HOW TO READ THIS DOCUMENT

This document is written in two layers throughout every section:

- **Black text** — the full technical content, written for physicists and researchers.
- ***Blue italic text*** *(labelled "In Plain English:")* — a jargon-free translation of the same idea, written for everyone else.

If you are not a physicist, you can read only the blue sections and still follow the entire theory from start to finish. Both layers are scientifically honest. Nothing has been dumbed down to the point of being wrong.

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Section 1 — The Big Idea

Modern physics rests on two pillars — General Relativity (GR), which describes gravity and the curvature of spacetime, and the Standard Model (SM), which describes particle physics and quantum fields. Both are confirmed by thousands of experiments to astonishing precision. Yet they are mathematically incompatible at the Planck scale (~1.616×10^{-35} m). No existing theory bridges this gap without invoking unobservable extra dimensions, untested supersymmetric partners, or strings at energies 15 orders of magnitude beyond experimental reach.

Unified Substrate Theory (UST) proposes a different path. A single rank-2 symmetric tensor substrate field $S_{\mu\nu}$ defined on a 4-dimensional Lorentzian manifold underlies all known physics. All forces emerge from a natural decomposition: $S_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} + \Phi_{\mu\nu}$, where $\eta_{\mu\nu}$ is flat spacetime, $h_{\mu\nu}$ encodes gravity, and $\Phi_{\mu\nu}$ encodes gauge fields and matter. Gravity, electromagnetism, and the nuclear forces are four different expressions of a single underlying structure.

UST has zero free parameters. Every coupling constant is fixed by a quantity already measured. The theory is not a fit to data — it is a derivation from first principles.

In Plain English: Physics has two rulebooks that contradict each other. One explains gravity. The other explains everything else — light, atoms, and the particles inside them. Both rulebooks work perfectly on their own, but no one has ever written one rulebook that covers everything. UST is an attempt at that single rulebook. The core idea is that the universe is not empty space with particles in it — it is a physical medium, like a sheet, and everything we experience is a vibration in that sheet. The sheet is the substrate field. Ripples in it become gravity. Oscillations in it become light. Distortions in it become particles. And critically, there are no adjustable dials. Every number in the theory is locked to something already measured in a laboratory.

Section 2 — The Substrate Field

The substrate field $S_{\mu\nu}$ is a rank-2 symmetric tensor — the same mathematical object Einstein used to describe spacetime curvature. It is defined at every point in 4-dimensional spacetime.

The field decomposes as:

$$S_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} + \Phi_{\mu\nu}$$

- $\boldsymbol{\eta_{\mu\nu}}$ — the flat Minkowski metric. This is empty spacetime. The background.
- $\mathbf{h_{\mu\nu}}$ — gravitational perturbation. Small deviations from flat spacetime. In the weak-field, slow-motion limit this recovers Newton's law of gravity and Einstein's field equations exactly.
- $\boldsymbol{\Phi_{\mu\nu}}$ — oscillatory modes. These are the wave-like excitations that produce electromagnetic fields and, through the SU(2) × SU(3) gauge structure, the weak and strong nuclear forces.

The single action from which all field equations are derived:

$$S_{UST} = \int d^4x \sqrt{(-g)}\ [\ \alpha R(g) + (1/4)\ g_S^2\ G^{\mu\nu}\ G_{\mu\nu} + L_{matter}(S_{\mu\nu}, \psi)\ ]$$

- $\boldsymbol{\alpha}$ = gravitational coupling constant, fixed by Newton's G.
- $\mathbf{g_S}$ = electromagnetic coupling constant, fixed by the fine structure constant.
- **R(g)** = Ricci scalar (Einstein's measure of curvature).
- $\mathbf{G^{\mu\nu}}$ = substrate field strength tensor.
- $\mathbf{L_{matter}}$ = matter Lagrangian.

Varying this action produces Einstein's equations in the $h_{\mu\nu}$ sector and a gauge field wave equation in the $\Phi_{\mu\nu}$ sector. Both from one equation.

In Plain English: Think of the substrate field as a mattress. A mattress has structure — springs, foam, fabric. Gravity is what happens when something heavy sits on the mattress and creates a depression. Light and the other forces are what happens when something vibrates the springs. UST *says there is one mattress, and everything — gravity, light, nuclear forces — is just a different kind of disturbance in it. The action is the single master equation that governs all of it, the way F = ma governs all of classical mechanics. Everything else is a consequence.*

Section 3 — Key Parameters: What We Know

UST has two fundamental coupling constants, both fixed by measured physical quantities.

Gravitational coupling:

$$\alpha = c^4 / (16\pi G) = 2.408 \times 10^{42}\ kg/s^2$$

Fixed by Newton's gravitational constant $G = 6.674 \times 10^{-11}\ \mathrm{m^3\ kg^{-1}\ s^{-2}}$. This is defined to exactly recover Newtonian gravity $\nabla^2\Phi = 4\pi G\rho$ in the weak-field limit.

Electromagnetic coupling:

$$g_S = \sqrt{(4\pi\ \alpha_{EM})} = \sqrt{(4\pi / 137.036)} = 0.302822$$

Fixed by the fine structure constant $\alpha_{EM} = 1/137.036$, the most precisely measured constant in physics.

These two numbers — and only these two — define the theory. The substrate particle mass M_S is not a free parameter. It is derived from the Fermilab muon g-2 measurement (Section 4). All other quantities are either constrained by experiment or follow from the above.

Complete parameter table:

- $\alpha = 2.408 \times 10^{42}$ kg/s² — fixed by Newton's G
- $g_S = 0.302822$ — fixed by $\alpha_{EM} = 1/137.036$
- $M_S = 58.9 \pm 3.5$ GeV — derived from Fermilab Δa_μ
- $R > 0.0124$ fm — lower bound from CCFR (binding constraint)
- $g_{res} < 0.010$ — upper bound from CCFR
- $\mu_s < 15.9$ GeV — scalar mass upper bound from $\hbar c/R$

- $\xi_S > 0.80$ AU — coherence length lower bound from Micius satellite Bell test

In Plain English: Most theories in physics have knobs you can turn. You adjust the knobs until the theory matches the data. UST has no knobs. The gravitational strength is set by Newton's G — a number measured in labs since the 1700s. The electromagnetic strength is set by the fine structure constant — a number measured so precisely it fills 12 decimal places. Once those two numbers go in, the rest of the theory is fixed. The mass of the new particle it predicts is not chosen — it falls out of the math automatically, locked to a measurement made at Fermilab.

Section 4 — The Soliton: A Particle That Is Also a Wave

The substrate particle is not a point particle. It is a topologically stable Gaussian soliton — a spatially extended, self-reinforcing wave packet:

$$\varphi_S(r) = A \times \exp(-r^2 / 2R^2)$$

R is the soliton radius, experimentally bounded to $R > 0.0124$ fm (femtometers), comparable to the proton radius.

The momentum-space form factor is the Fourier transform of this profile:

$$F(q) = \exp(-q^2R^2/4)$$

q = momentum transfer. This form factor creates two distinct coupling regimes:

- At **low momentum** ($q \sim m_\mu = 0.106$ GeV, the muon g-2 scale): $F(q) \approx 1$. Full coupling g_S applies. UST produces the observed muon anomaly.
- At **high momentum** ($q \sim M_S = 59$ GeV, the collider scale): F(q) is exponentially suppressed. The substrate particle is effectively invisible to collider detectors.

This is directly analogous to the proton electromagnetic form factor in QCD. A proton has unit electric charge — but probe it at short wavelengths and the effective charge appears to smear out because you are resolving its internal structure. The substrate soliton works identically. The form factor is not an ad hoc addition. It is the mathematical consequence of the soliton ansatz.

The Muon Prediction — The Key Result

The Fermilab Muon g-2 experiment measures the muon anomalous magnetic moment a_μ. The discrepancy with the Standard Model is:

$$\Delta a_\mu = (249 \pm 48) \times 10^{-11} \quad (5.2\sigma)$$

UST's one-loop contribution at the muon scale (where F ≈ 1):

$$\delta a_\mu = g_S^2 \times m_\mu^2 / (12\pi^2 \times M_S^2)$$

Solving for M_S:

$$M_S = g_S \times m_\mu / \sqrt{(12\pi^2 \times \Delta a_\mu)} = 58.9\ ^{+3.4}/_{-3.5}\ GeV$$

Verification: $\delta a_\mu = (0.302822)^2 \times (0.10566)^2 / (12\pi^2 \times (58.9)^2) = 249 \times 10^{-11}$. Exact match.

Exact One-Loop Integral Validation

$$\delta a_\mu\ (exact) = (g_S^2/4\pi^2) \times \int_0^1 dx\ [x^2(1-x)] / [x^2 + (M_S/m_\mu)^2(1-x)]$$

Numerical result: 249.138×10^{-11}. Approximation: 249.164×10^{-11}. Error: 0.0102%. M_S shift: 3 MeV — four orders of magnitude below experimental uncertainty of ±3,400 MeV. The heavy-mass approximation is validated.

In Plain English: The substrate particle is not a tiny dot — it is a blob with a definite size, like a droplet of water rather than a grain of sand. That size is the key to everything. When you prod the particle gently — the way the muon experiment does — you feel its full weight. When you smash into it hard — the way the Large Hadron Collider does — it feels like almost nothing, because you are hitting it so fast that it effectively blurs past you. This is exactly how the proton behaves. Hit a proton gently and it feels like a solid charged ball. Hit it at LHC energies and it feels like it is mostly empty space. The substrate particle works the same way. That is why Fermilab sees its effect on the muon, but CERN has not spotted the particle itself. The prediction: a new particle at 58.9 billion electron volts. That number came from plugging in two measured constants and solving one equation. No guessing. No tuning.

Section 5 — Four Testable Predictions

UST makes four independently falsifiable predictions. Any one of them could rule out the theory.

P-001 — Substrate Resonance at 58.9 GeV. A neutral resonance at $M_S = 58.9\ ^{+3.4}/_{-3.5}$ GeV detectable as a cross-section enhancement in dilepton or diphoton final states at a future e^+e^- collider with $\sqrt{s} \sim 60$ GeV. Target experiments: FCC-ee, CEPC. A non-observation with sufficient luminosity at this energy would falsify UST.

P-002 — Bell Inequality Distance Correction. UST predicts a systematic distance-dependent correction to entanglement fidelity: $\delta E/E = r/\xi_S$, where $\xi_S > 0.80$ AU (constrained by the Micius satellite Bell test at 1,203 km). Below current sensitivity at satellite baselines. Falsifiable at next-generation long-baseline quantum entanglement experiments.

P-003 — Muon Anomalous Magnetic Moment. $\delta a_\mu = g_S^2\, m_\mu^2 / (12\pi^2 M_S^2) = 249 \times 10^{-11}$. Already consistent with Fermilab Runs 1–3 $(249 \pm 48) \times 10^{-11}$. Runs 4–6 will reduce uncertainty. A revised central value shifting more than ~3σ from 249×10^{-11} would falsify or significantly constrain UST.

P-004 — Light Neutral Scalar Below 15.9 GeV. The soliton sector predicts a spin-0, electrically neutral scalar with mass $\mu_s < \hbar c/R < 15.9$ GeV and coupling $g_{scalar} \ll 0.01$. Observable channels: $B \to K$ + missing energy at Belle II; $K^+ \to \pi^+$ + invisible at NA62; electron beam dump at NA64. All three are currently operating.

In Plain English: A theory that cannot be proven wrong is not science — it is philosophy. UST makes four specific bets. Bet one: a new particle exists at roughly 59 GeV. Future electron-positron colliders will either find it or they won't. Bet two: pairs of quantum-entangled particles separated by very large distances will show a tiny weakening of their connection — a distance penalty. Bet three: the muon measurement at Fermilab is real and will hold up as more data comes in. Bet four: a second, lighter particle exists below 15.9 GeV, so lightweight it slips through most detectors unnoticed. Three experiments running right now are sensitive to exactly this kind of ghost particle.

Section 6 — Why Current Colliders Haven't Seen It Yet

Three independent experimental constraints have been applied. All are satisfied.

CMS EXO-19-019 (LHC Run 2, 140 fb^{-1}, 13 TeV dilepton search): The CMS search covers invariant mass $m_{\ell\ell}$ > 120 GeV. M_S = 58.9 GeV is below this threshold. UST is not constrained by this search. Status: Not applicable.

LEP2 Electroweak Precision ($\sqrt{s}$ = 130–209 GeV, ~1% precision): Without the form factor: 219% correction to electroweak observables — excluded. With $F(q) = \exp(-q^2R^2/4)$ at LEP2 momentum scales, correction drops to sub-percent. Requires R > 0.011 fm. Status: Satisfied.

CCFR Neutrino Trident Production: On-shell residual coupling must satisfy $g_{res} < 0.010$. Via the form factor, this requires R > 0.0124 fm. This is the binding constraint. Status: Satisfied.

BMW Lattice QCD Robustness

Under all three HVP scenarios, UST survives. As M_S increases, form factor suppression at collider scales strengthens, improving consistency with null collider results.

Table 1. UST substrate mass under three HVP scenarios.

HVP Scenario	**Δa_μ ($\times 10^{-11}$)**	**Significance**	**M_S (GeV)**	**Below CMS 120 GeV?**
Data-driven (primary)	249	5.2σ	58.9	Yes
CMD-3 blend	177	3.6σ	69.9	Yes
BMW lattice QCD	105	2.2σ	90.7	Yes

In Plain English: The obvious question is: if this particle exists, why hasn't the LHC found it? Three reasons. First, the LHC's particle searches in the relevant channel only kick in above 120 GeV. The substrate particle sits at 59 GeV — below the radar. Second, the particle's blob-like shape means it barely registers at LHC collision energies — like trying to catch a soap bubble by throwing a baseball at it. Third, even if there is an ongoing argument about the exact size of the muon anomaly, UST survives every version of that argument. The predicted particle mass changes *a little — 59, 70, or 91 GeV depending on which calculation you trust — but it stays below the LHC's detection threshold in every case.*

Section 7 — Five Self-Consistency Checks (All Pass)

UST must satisfy internal consistency requirements independent of experimental constraints.

Check 1 — Dimensional consistency. All terms in L_{UST} carry dimension $[\text{Energy}]^4$ in natural units. The couplings α and g_S have dimensions that make each term in the action dimensionally uniform. Pass.

Check 2 — Newtonian limit. In the weak-field slow-motion limit, the $h_{\mu\nu}$ sector of the UST field equations reduces to $\nabla^2\Phi = 4\pi G\rho$ with coefficient $G = c^4/(16\pi\alpha)$. The gravitational coupling is defined precisely to enforce this. Pass.

Check 3 — Propagation speed. Linearizing the substrate field equations yields dispersion relations $\omega^2 = k^2c^2$ for both gravitational and electromagnetic wave modes. Both propagate at exactly c in vacuum. Consistent with gravitational wave arrival time constraints from GW170817/GRB170817A. Pass.

Check 4 — Heisenberg uncertainty principle. Canonical quantization of the $\Phi_{\mu\nu}$ sector yields commutation relations $[\varphi_S(x), \pi_S(y)] = i\hbar\delta^3(x-y)$. From this, $\Delta x\Delta p \geq \hbar/2$ follows by standard operator algebra. UST does not modify quantum mechanics. Pass.

Check 5 — Energy-momentum conservation. The contracted Bianchi identity $\nabla_\mu G^{\mu\nu} = 0$ applied to the UST field equations implies $\nabla_\mu T^{\mu\nu} = 0$. Local energy-momentum conservation is a consequence of diffeomorphism invariance, not an additional postulate. Pass.

In Plain English: Before a theory gets to make predictions, it has to pass basic sanity checks. It can't violate rules that every other working theory obeys. UST passes five of them. The units work out. In everyday slow-speed weak-gravity conditions, it reproduces Newton's law exactly. Gravitational waves and light both travel at the speed of light — as observed. It does not tamper with the uncertainty principle or break energy conservation. These are not optional — a theory *that fails any one of them is automatically wrong, regardless of what it predicts. UST passes all five.*

Section 8 — What's Still Unknown

Three open problems are identified. None affect the four falsifiable predictions at current experimental precision.

Open problem 1 — Soliton self-coupling γ. The Gaussian soliton ansatz requires a self-coupling term to be topologically stable. A dimensional estimate gives $\gamma \sim 10^{33}$–10^{37}, but the precise value requires high-performance numerical simulation. This parameter does not enter any of the four predictions at one-loop order.

Open problem 2 — Full second quantization. The current treatment quantizes the linearized substrate field. A complete quantum field theory of $S_{\mu\nu}$ — with Feynman rules, renormalization group equations, and running couplings — is in progress. Preliminary analysis suggests the theory is renormalizable by power counting, but this has not been proven.

Open problem 3 — Explicit SU(2) × SU(3) derivation. The $\Phi_{\mu\nu}$ sector is asserted to reproduce the weak and strong gauge groups. A complete derivation from the UST action to $SU(2)_L \times U(1)_Y$ and $SU(3)_c$ — including symmetry breaking, the Higgs mechanism, and quark confinement — is reserved for follow-up papers.

In Plain English: UST is not finished. Three things still need to be worked out. First, the exact strength of the substrate field's self-interaction — how strongly it pulls itself together to stay as a stable blob — needs a full computer simulation to pin down. Second, the full quantum version of the theory — with all the machinery needed to compute particle collision probabilities from scratch — is still being built. Third, UST currently asserts that the weak and strong nuclear forces emerge from the substrate field, but hasn't yet spelled out every step of how that happens. None of these gaps affect the testable predictions in Section 5. They are the next three papers.

Section 9 — How to Test This: What Equipment We Need

The four predictions map onto specific experimental programs.

P-001 (Substrate resonance at 58.9 GeV): Required: e^+e^- collider with $\sqrt{s}$ tunable from 55–65 GeV and sufficient luminosity to detect a narrow resonance in dilepton or diphoton channels. Target experiments: FCC-ee (CERN, planned), CEPC (China, planned). Timeline: 2035–2045. Observable: Cross-section enhancement $\sigma(e^+e^- \to \ell^+\ell^-$ or $\gamma\gamma)$ above QED prediction at $\sqrt{s} \approx 59$ GeV.

P-002 (Entanglement distance correction): Required: Long-baseline Bell test with baseline $r \gg$ 1,203 km and entanglement fidelity precision better than r/ξ_S. Target: Next-generation quantum satellite experiments, deep-space Bell tests. Observable: Systematic fidelity reduction scaling linearly with baseline distance.

P-003 (Muon g-2): Required: Continued operation of Fermilab Muon g-2 experiment (Runs 4–6 in progress). Observable: Confirmation that the anomaly Δa_μ remains consistent with (249 ± smaller uncertainty) × 10^{-11}.

P-004 (Light scalar below 15.9 GeV): Required: Experiments sensitive to missing energy signatures from light neutral particles. Target experiments: Belle II (KEK, Japan) — $B \to K + E_{miss}$; NA62 (CERN) — $K^+ \to \pi^+$ + invisible; NA64 (CERN) — electron beam dump, missing energy. All three are currently taking data.

In Plain English: Prediction 1 needs a new collider — a gentler, more precise machine than the LHC that can be tuned to exactly 59 GeV. Two are being planned. It will take roughly a decade to build. Prediction 2 needs a future satellite that can run quantum entanglement tests over much longer distances than currently possible. Predictions 3 and 4 can be tested right now with equipment that already exists. The Fermilab muon experiment is still running. The Belle II, NA62, and NA64 experiments are already collecting the data needed to find or rule out the light scalar. UST is not waiting on technology that doesn't exist — three of its four predictions are testable today.

Section 10 — Why This Matters

The significance of UST, if validated, extends across multiple domains.

Theoretical physics: A successful zero-parameter unification would resolve the fundamental incompatibility between GR and the Standard Model without invoking extra dimensions, supersymmetry, or strings. It would establish that the four known forces are not independent — they are a single substrate field observed under different conditions.

Experimental physics: The predicted substrate resonance at 58.9 GeV gives the FCC-ee and CEPC programs a concrete, quantitative target. Rather than scanning a wide parameter space, experimenters have a specific mass, decay channel, and coupling strength to look for.

Quantum information: The predicted coherence length $\xi_S > 0.80$ AU would, if confirmed, represent the first measurement of a property of the substrate medium itself — a direct probe of the structure of the vacuum.

Philosophy of science: UST is fully falsifiable. It makes specific, quantitative predictions that will either be confirmed or excluded by experiments planned within the next two decades. This places it in a different category from other unification proposals that operate at energies permanently beyond experimental reach.

In Plain English: If UST is right, the payoff is enormous. It would mean that gravity, light, and the nuclear forces are not four separate things that physicists have to keep track of with four different rulebooks. They are one thing — four faces of a single substrate. That would be the deepest simplification in the history of physics. On a practical level, it gives the people building the next generation of particle colliders a very specific shopping list: go to 59 GeV and look for a neutral resonance. On a philosophical level, UST is science in the classical sense — it sticks its neck out. It can be wrong. It will know within twenty years whether it is. That is what makes it physics rather than speculation.

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